

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.2 – 1.4 Vectors Worksheet

1. Let $\mathbf{u} = \langle 1, 0, 2 \rangle$, $\mathbf{v} = \langle -3, 1, 1 \rangle$, and $\mathbf{w} = \langle 2, -1, 1 \rangle$ be vectors in $\mathbb{R}^3$.

a) Determine the exact value of $\|2\mathbf{u} - \mathbf{v}\|$.

We calculate that $2\mathbf{u} - \mathbf{v} = \langle 2, 0, 4 \rangle - \langle -3, 1, 1 \rangle = \langle 5, -1, 3 \rangle$ and therefore,

$$\|2\mathbf{u} - \mathbf{v}\| = \sqrt{25 + 1 + 9} = \sqrt{35}$$

b) Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Determine the exact value of θ , then determine whether $0 < \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta < \pi$.

We know that $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos(\theta) \rightarrow \theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right)$. Therefore, we calculate that

$$\theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right) = \cos^{-1}\left(\frac{-3 + 0 + 2}{\sqrt{5}\sqrt{11}}\right) = \cos^{-1}\left(\frac{-1}{\sqrt{55}}\right)$$

Since we can see that $\cos(\theta) = -\frac{1}{\sqrt{55}}$ this implies that θ would have to be an obtuse angle and thus

$$\frac{\pi}{2} < \theta < \pi.$$

c) Mark the answer that best describes the meaning of the expression $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$.

☐

It is the sum of the areas of the two parallelograms determined by the two pairs $\{\mathbf{u}, \mathbf{v}\}$ and $\{\mathbf{u}, \mathbf{w}\}$.

☐

It is the area of the parallelogram with vertices determined by 0 , $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

☒

It is the volume of the parallelepiped determined by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

☐

It has no meaning, but it is always defined and sometimes it is zero.

☐

It is undefined.

2. Consider the points in the plane shown at the right.

• A

a) Determine the name of a vector that is equal to $2\overrightarrow{DE} - \overrightarrow{CA}$.

• B

We can see that we can represent $2\overrightarrow{DE}$ as $\overrightarrow{CE}$ and represent $-\overrightarrow{CA}$ as $\overrightarrow{AC}$. Therefore, we can conclude that $2\overrightarrow{DE} - \overrightarrow{CA} = \overrightarrow{AE}$. The only vector that has the same direction and magnitude as $\overrightarrow{AE}$ is $\overrightarrow{DG}$. See the diagram below.

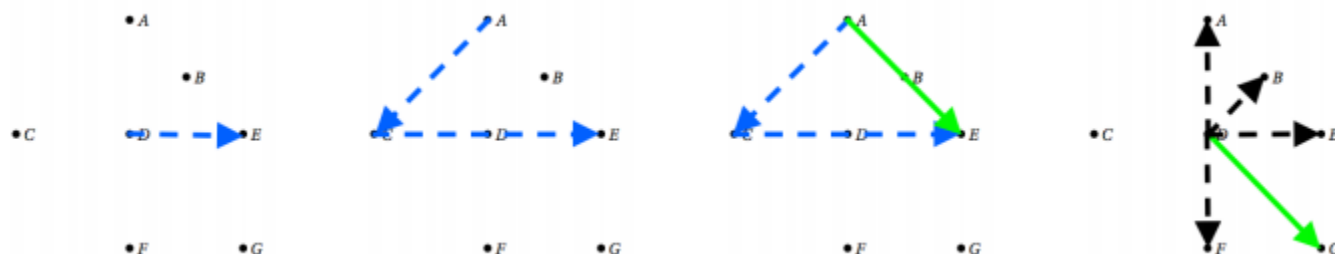
• C

• D

• E

• F

• G



b) Determine the name of a vector that is equal to $\text{proj}_{\overrightarrow{CA}}(\overrightarrow{DE})$.

Once we have these two vectors pictured we can move the vector $\overrightarrow{CA}$ so that it begins at point D. Now we can more easily see how the projection will work. We draw out our right triangle and note that the projection will be represented as the vector $\overrightarrow{DB}$. See the diagram below.



3. Determine a unit vector which has the opposite direction of $\mathbf{w} = \langle 4, 3, 1 \rangle$.

We see that $\|\mathbf{w}\| = \sqrt{16+9+1} = \sqrt{26}$ and so a unit vector that points in the *same* direction as $\mathbf{w}$ would be

$\mathbf{u} = \left\langle \frac{4}{\sqrt{26}}, \frac{3}{\sqrt{26}}, \frac{1}{\sqrt{26}} \right\rangle$. Therefore, a unit vector that points in the *opposite* direction as $\mathbf{w}$ would be

$-\mathbf{u} = \left\langle -\frac{4}{\sqrt{26}}, -\frac{3}{\sqrt{26}}, -\frac{1}{\sqrt{26}} \right\rangle$.

4. Consider the three points $A = (0, -1, 1)$, $B = (1, -1, 3)$, and $C = (2, 0, 0)$ in $\mathbb{R}^3$.

a) Find $\mathbf{v} = \overrightarrow{AB}$ and $\mathbf{w} = \overrightarrow{BC}$.

We see that $\mathbf{v} = \langle 1, 0, 2 \rangle$ and $\mathbf{w} = \langle 1, 1, -3 \rangle$.

b) Calculate the cross-product $\mathbf{v} \times \mathbf{w}$.

$$\text{We see that } \mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 1 & 1 & -3 \end{vmatrix} = \langle 0 - 2, -(-3 - 2), 1 - 0 \rangle = \langle -2, 5, 1 \rangle.$$

c) Find the area of the parallelogram determined by $\mathbf{v}$ and $\mathbf{w}$.

The area of the parallelogram is given by $\text{Area} = \|\mathbf{v} \times \mathbf{w}\| = \sqrt{4 + 25 + 1} = \sqrt{30}$

5. Determine whether the vectors $\mathbf{a} = \langle -2, 5, 3 \rangle$ and $\mathbf{b} = \langle 2, 2, -2 \rangle$ are orthogonal, parallel, or neither.

We quickly calculate that $\mathbf{a} \cdot \mathbf{b} = -4 + 10 - 6 = 0$. This implies that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

6. Determine whether the vectors $\mathbf{a} = \langle -2, 3, 4 \rangle$ and $\mathbf{b} = \langle 1, -3, 2 \rangle$ are orthogonal, parallel, or neither.

We calculate $\mathbf{a} \cdot \mathbf{b} = -2 - 9 + 8 = -3 \neq 0$ so these vectors are not orthogonal.

We calculate $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & 4 \\ 1 & -3 & 2 \end{vmatrix} = \langle 6 - (-12), -(-4 - 4), 6 - 3 \rangle = \langle 18, 8, 3 \rangle \neq \mathbf{0}$ so these vectors are not parallel.

Therefore, these vectors are neither orthogonal nor parallel.

7. Consider the three points $A = (0, 1, 0)$, $B = (1, 1, 1)$, and $C = (0, 2, 1)$ in $\mathbb{R}^3$.

a) Determine the projection of the vector $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

We see that $\overrightarrow{AC} = \langle 0, 1, 1 \rangle$ and $\overrightarrow{AB} = \langle 1, 0, 1 \rangle$. Therefore,

$$\text{proj}_{\overrightarrow{AB}}(\overrightarrow{AC}) = \left(\frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{\|\overrightarrow{AB}\|^2} \right) \overrightarrow{AB} = \left(\frac{0+0+1}{(\sqrt{2})^2} \right) \langle 1, 0, 1 \rangle = \frac{1}{2} \langle 1, 0, 1 \rangle = \left\langle \frac{1}{2}, 0, \frac{1}{2} \right\rangle$$

b) Determine the area of the triangle formed by these three points.

We see that $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = \langle 0-1, -(1-0), 1-0 \rangle = \langle -1, -1, 1 \rangle$. So, the area of the triangle is given

$$\text{by } Area = \frac{1}{2} \|\overrightarrow{AB} \times \overrightarrow{AC}\| = \frac{1}{2} (\sqrt{3}) = \frac{\sqrt{3}}{2}$$

c) Calculate $\overrightarrow{AB} \cdot (\overrightarrow{AB} \times \overrightarrow{AC})$.

We know from part (b) that $\overrightarrow{AB} \times \overrightarrow{AC} = \langle -1, -1, 1 \rangle$ and so $\overrightarrow{AB} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) = -1+0+1=0$.

Another way of doing this would be to recognize that

$$\overrightarrow{AB} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) = (\overrightarrow{AB} \cdot \overrightarrow{AB}) \times (\overrightarrow{AB} \cdot \overrightarrow{AC}) = 0 \times (\overrightarrow{AB} \cdot \overrightarrow{AC}) = 0$$

d) For the vector $\mathbf{v} = \langle 0, 2, 2 \rangle$, determine the angle between $\overrightarrow{AC}$ and $\mathbf{v}$.

We know that $\overrightarrow{AC} \cdot \mathbf{v} = \|\overrightarrow{AC}\| \|\mathbf{v}\| \cos(\theta) \rightarrow \theta = \cos^{-1} \left(\frac{\overrightarrow{AC} \cdot \mathbf{v}}{\|\overrightarrow{AC}\| \|\mathbf{v}\|} \right)$. Therefore, we calculate that

$$\theta = \cos^{-1} \left(\frac{\overrightarrow{AC} \cdot \mathbf{v}}{\|\overrightarrow{AC}\| \|\mathbf{v}\|} \right) = \cos^{-1} \left(\frac{0+2+2}{\sqrt{2} \sqrt{8}} \right) = \cos^{-1} \left(\frac{4}{\sqrt{16}} \right) = \cos^{-1}(1) = 0$$

Another way of doing this would be to simply recognize that $\mathbf{v} = 2\overrightarrow{AC}$ and therefore they point in the same direction and thus the angle between these vectors must be $\theta = 0$.

8. Consider the vectors **a**, **b**, and **c** shown at the right.

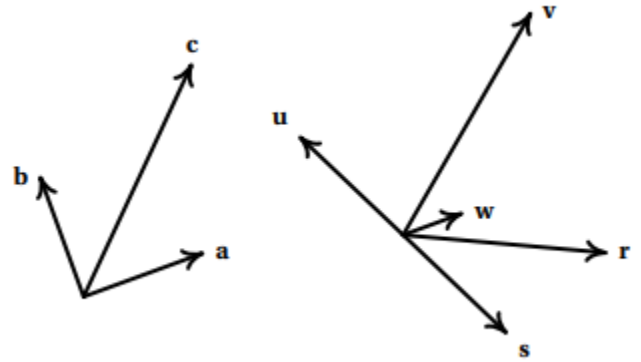
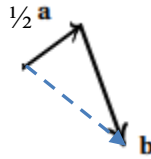
Their lengths are $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$ and $\|\mathbf{c}\| = 2$.

- a) Circle the vector that best represents $\frac{1}{2}\mathbf{a} - \mathbf{b}$.



We can re-express this as $\frac{1}{2}\mathbf{a} - \mathbf{b} = \frac{1}{2}\mathbf{a} + (-\mathbf{b})$.

Then, using basic vector addition properties we know that the sum should look like the dashed vector in blue. Therefore, we conclude that **s** is the best choice here.



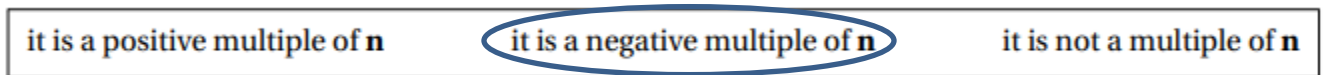
- b) Circle the number closest to $\mathbf{a} \cdot \mathbf{c}$.



We know that $\mathbf{a} \cdot \mathbf{c} = \|\mathbf{a}\|\|\mathbf{c}\|\cos(\theta)$ where θ is the angle between the vectors **a** and **c**. Based on the given information we know that $\mathbf{a} \cdot \mathbf{c} = 1 \cdot 2 \cos(\theta) = 2 \cos(\theta)$ and since from the picture we see that $\theta \approx \frac{\pi}{4}$ we

know that $\cos(\theta) \approx \frac{\sqrt{2}}{2}$ and so $\mathbf{a} \cdot \mathbf{c} = 2 \cos(\theta) = 2 \left(\frac{\sqrt{2}}{2} \right) = \sqrt{2}$.

- c) A nonzero vector **n** is pointing directly up out of the page. Circle the best description of $\mathbf{b} \times \mathbf{c}$.



We can use the right-hand rule to determine the direction of $\mathbf{b} \times \mathbf{c}$. When we do this, we see that $\mathbf{b} \times \mathbf{c}$ would point *into* the page, which is in the opposite direction of **n**. Therefore, $\mathbf{b} \times \mathbf{c}$ is a negative multiple of **n**.

9. Determine all values of a for which the vectors $\langle a, -3, 1 \rangle$ and $\langle a, 2a, -7 \rangle$ are perpendicular.

In order for these vectors to be perpendicular you we must have:

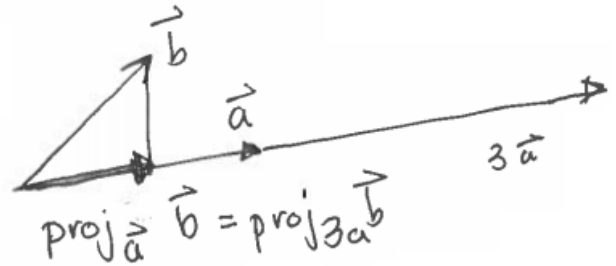
$$\begin{aligned} \langle a, -3, 1 \rangle \cdot \langle a, 2a, -7 \rangle &= 0 \rightarrow a^2 - 6a - 7 = 0 \\ &\rightarrow (a - 7)(a + 1) = 0 \\ &\rightarrow a = 7 \text{ or } a = -1 \end{aligned}$$

10. Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors in $\mathbb{R}^3$ such that $\|\text{proj}_{\mathbf{a}}(\mathbf{b})\| = 2$.

a) Determine the value of $\|\text{proj}_{3\mathbf{a}}(\mathbf{b})\|$.

We can see that if we project $\mathbf{b}$ onto $3\mathbf{a}$ our projection vector isn't any different and therefore we would conclude that $\|\text{proj}_{3\mathbf{a}}(\mathbf{b})\| = 2$.

We could also look at this in the following way



$$\text{proj}_{3\mathbf{a}}(\mathbf{b}) = \left(\frac{3\mathbf{a} \cdot \mathbf{b}}{\|3\mathbf{a}\|^2} \right) 3\mathbf{a} = 9 \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|3\mathbf{a}\|^2} \right) \mathbf{a} = 9 \left(\frac{\mathbf{a} \cdot \mathbf{b}}{3^2 \|\mathbf{a}\|^2} \right) \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = \text{proj}_{\mathbf{a}}(\mathbf{b})$$

b) Determine the value of $\|\text{proj}_{\mathbf{a}}(5\mathbf{b})\|$.

$$\text{proj}_{\mathbf{a}}(5\mathbf{b}) = \left(\frac{\mathbf{a} \cdot 5\mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = 5 \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = 5 \text{proj}_{\mathbf{a}}(\mathbf{b})$$

$$\text{Therefore, } \|\text{proj}_{\mathbf{a}}(5\mathbf{b})\| = \|5 \text{proj}_{\mathbf{a}}(\mathbf{b})\| = 5 \|\text{proj}_{\mathbf{a}}(\mathbf{b})\| = 5(2) = 10.$$

11. Determine all values of a and b for which $\langle a, -3, 2b \rangle$ and $\langle a, a, 1 \rangle$ are parallel.

In order for these vectors to be parallel they must be scalar multiples of each other. Therefore, it must be the case that

$$\begin{aligned} \lambda \langle a, -3, 2b \rangle &= \langle a, a, 1 \rangle \rightarrow \langle \lambda a, -3\lambda, 2\lambda b \rangle = \langle a, a, 1 \rangle \\ &\rightarrow \begin{cases} \lambda a = a \\ -3\lambda = a \\ 2\lambda b = 1 \end{cases} \end{aligned}$$

In order for the first equation to be true we must either have $a = 0$ or $\lambda = 1$. We investigate each case separately.

- If $a = 0$ then by the 2nd equation we must have that $\lambda = 0$. However, this then implies, by the 3rd equation that $0 = 1$. Therefore, it cannot be the case that $a = 0$.
- If $\lambda = 1$ then by the 2nd equation we must have that $a = -3$ and by the 3rd equation $b = \frac{1}{2}$.

Therefore, the only values of a and b for which the vectors are parallel are $a = -3$ and $b = \frac{1}{2}$.